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BATCH -16

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EXPERIMENT-5

1. C Program to Find Roots of a Quadratic Equation

   Aim

   Write a C program to find the roots of a quadratic equation of the form:

$$ax^2 + bx + c = 0$$

   Input the coefficients a, b, and c, where $a \neq 0$.

   Calculate the discriminant:

$$D = b^2 - 4ac$$

   The roots are determined according to the value of D.

   Conditions

Condition	Output
$D > 0$	Two real and distinct roots
$D = 0$	Two real and equal roots
$D < 0$	Two imaginary/complex roots

   For $D > 0$:

$$root1 = \frac{-b + \sqrt{D}}{2a}$$

$$root2 = \frac{-b - \sqrt{D}}{2a}$$

   For $D = 0$:

$$root1 = root2 = \frac{-b}{2a}$$

For D < 0:

$$Real\ Part = \frac{-b}{2a}$$

$$Imaginary\ Part = \frac{\sqrt{-D}}{2a}$$

<u>Algorithm</u>

1. Start.
2. Declare variables a, b, c, D, root1, root2, realPart, and imagPart.
3. Input the values of a, b, and c.
4. Check whether a = 0.
5. If a = 0, display that it is not a quadratic equation.
6. Otherwise, calculate:
   D = b*b - 4*a*c.
7. If D > 0:
   - Calculate two real roots.
   - Display both roots.
8. Else if D = 0:
   - Calculate the repeated root.
   - Display both equal roots.
9. Else:
   - Calculate the real and imaginary parts.
   - Display the two complex roots.
10. Stop.

<u>Pseudocode</u>

START

INPUT a, b, c

```
IF a = 0 THEN
   DISPLAY "Not a quadratic equation"
ELSE
   D = b² - 4ac

   IF D > 0 THEN
      root1 = (-b + √D) / (2a)
      root2 = (-b - √D) / (2a)
      DISPLAY "Two real and distinct roots"
      DISPLAY root1, root2

   ELSE IF D = 0 THEN
      root1 = -b / (2a)
      root2 = root1
      DISPLAY "Two real and equal roots"
      DISPLAY root1, root2

   ELSE
      realPart = -b / (2a)
      imagPart = √(-D) / (2a)
      DISPLAY "Two imaginary/complex roots"
      DISPLAY realPart + imagPart i
      DISPLAY realPart - imagPart i
   END IF
END IF

STOP
```

Test Case Table

Test Case	a	b	c	D	Expected Output
TC01	1	-5	6	1	Two real distinct roots: 3, 2
TC02	1	-4	4	0	Two real equal roots: 2, 2
TC03	1	2	5	-16	Complex roots: -1 + 2i, -1 - 2i
TC04	0	2	3	—	Not a quadratic equation

C Program

```
#include <stdio.h>
#include <math.h>

int main()
{
    double a, b, c, D;
    double root1, root2;
    double realPart, imagPart;

    printf("Enter coefficients a, b and c: ");
    scanf("%lf %lf %lf", &a, &b, &c);

    if (a == 0)
    {
        printf("Not a quadratic equation.\n");
    }
    else
    {
```

```c
D = b * b - 4 * a * c;

printf("Discriminant = %.2lf\n", D);

if (D > 0)
{
    root1 = (-b + sqrt(D)) / (2 * a);
    root2 = (-b - sqrt(D)) / (2 * a);

    printf("Two real and distinct roots.\n");
    printf("Root 1 = %.2lf\n", root1);
    printf("Root 2 = %.2lf\n", root2);
}
else if (D == 0)
{
    root1 = -b / (2 * a);
    root2 = root1;

    printf("Two real and equal roots.\n");
    printf("Root 1 = %.2lf\n", root1);
    printf("Root 2 = %.2lf\n", root2);
}
else
{
    realPart = -b / (2 * a);
    imagPart = sqrt(-D) / fabs(2 * a);

    printf("Two imaginary/complex roots.\n");
    printf("Root 1 = %.2lf + %.2lfi\n",
           realPart, imagPart);
    printf("Root 2 = %.2lf - %.2lfi\n",
           realPart, imagPart);
```

```
        }
    }

    return 0;
}
```

Compilation

```
gcc quadratic.c -o quadratic -lm
```

Execution

```
./quadratic
```

Sample Output — D > 0

```
Enter coefficients a, b and c: 1 -5 6
Discriminant = 1.00
Two real and distinct roots.
Root 1 = 3.00
Root 2 = 2.00
```

Sample Output — D = 0

```
Enter coefficients a, b and c: 1 -4 4
Discriminant = 0.00
Two real and equal roots.
Root 1 = 2.00
Root 2 = 2.00
```

Sample Output — D < 0

```
Enter coefficients a, b and c: 1 2 5
Discriminant = -16.00
Two imaginary/complex roots.
Root 1 = -1.00 + 2.00i
Root 2 = -1.00 - 2.00i
```

---

2. Menu-Driven C Program to Calculate Income Tax

Aim

Write a menu-driven C program to calculate income tax under the old and new tax regimes in India.

The program accepts:

- Taxpayer's name
- Age
- Gross income
- Eligible deductions

and calculates the final tax payable.

(a) Calculation of Tax in Old Regime

Input

The program accepts:

standard deduction

deduction savings

deduction pension

deduction medical

home loan interest

savings account interest

senior citizen interest

Maximum Deduction Limits

Deduction	Maximum Limit
Standard deduction	₹50,000
Deduction savings	₹1,50,000
Deduction pension	₹50,000
Deduction medical	₹25,000
Home loan interest	₹2,00,000
Savings account interest	₹10,000
Senior citizen interest	₹50,000

Each deduction should be restricted to its maximum permitted limit.

Calculate Total Deduction and Taxable Income

total_deductions = sum of allowed deductions

taxable_income = gross_income - total_deductions

If:

taxable_income < 0

then:

taxable_income = 0

Old Regime Tax Slabs

Taxable Income Slab	Non-Senior Citizen (<60 years)	Senior Citizen (60 to <80 years)	Super Senior Citizen (>=80 years)
Up to ₹2,50,000	Nil	Nil	Nil
₹2,50,001 to ₹3,00,000	5%	Nil	Nil
₹3,00,001 to ₹5,00,000	5%	5%	Nil
₹5,00,001 to ₹10,00,000	20%	20%	20%
Above ₹10,00,000	30%	30%	30%

Old Regime Tax Rebate

Taxable Income Limit	Rebate
≤ ₹5,00,000	Minimum of ₹12,500 and tax calculated above
> ₹5,00,000	₹0

(b) Calculation of Tax in New Regime

Note: The image labels this section as "(b) Calculation of Tax in Old regime", but the slab table shown is clearly the **new regime** slab structure. So it should be written as **New Regime.**

Calculate Taxable Income

Under the new regime:

taxable_income = gross_income

No deductions from the old-regime section are applied here.

New Regime Tax Slabs

Taxable Income Slab	Tax Rate
Up to ₹4,00,000	Nil
₹4,00,001 to ₹8,00,000	5%
₹8,00,001 to ₹12,00,000	10%
₹12,00,001 to ₹16,00,000	15%
₹16,00,001 to ₹20,00,000	20%
₹20,00,001 to ₹24,00,000	25%
Above ₹24,00,000	30%

New Regime Tax Rebate

Taxable Income Limit	Rebate
≤ ₹12,00,000	Minimum of ₹60,000 and tax calculated above
> ₹12,00,000	₹0

(c) Final Calculations

These calculations are common to both regimes.

tax_after_rebate = tax - rebate

health_education_cess = 4% of tax_after_rebate

final_tax = tax_after_rebate + health_education_cess

(d) Output

The program should display:

Name

Age

Gross Income

Taxable Income

Income Tax Before Rebate

Rebate

Income Tax After Rebate

Health & Education Cess

Final Tax Payable

Algorithm — Income Tax Program

1. Start.
2. Input taxpayer's name.
3. Input taxpayer's age.

4. Input gross income.
5. Display menu:
    - Old Tax Regime
    - New Tax Regime
    - Exit
6. If the user selects the old regime:
    - Input all deductions.
    - Apply the maximum limits to the deductions.
    - Calculate total deductions.
    - Calculate taxable income.
    - If taxable income is negative, set it to zero.
    - Determine the applicable tax slabs according to age.
    - Calculate income tax.
    - Calculate old-regime rebate.
7. If the user selects the new regime:
    - Set taxable income equal to gross income.
    - Calculate tax according to the new-regime slabs.
    - Calculate new-regime rebate.
8. Calculate tax after rebate.
9. Calculate 4% Health and Education Cess.
10. Calculate final tax payable.
11. Display all details.
12. If Exit is selected, terminate the program.
13. Stop.

Pseudocode

```
START

INPUT name
INPUT age
INPUT gross_income
```

```
DISPLAY MENU

1. Old Tax Regime
2. New Tax Regime
3. Exit

INPUT choice

IF choice = 1 THEN

    INPUT all deductions

    Limit each deduction according to maximum allowed limit

    total_deductions =
        standard deduction +
        deduction savings +
        deduction pension +
        deduction medical +
        home loan interest +
        savings account interest +
        senior citizen interest

    taxable_income = gross_income - total_deductions

    IF taxable_income < 0 THEN
        taxable_income = 0
    END IF

    Determine age category

    Calculate tax using old-regime slabs
```

```
    IF taxable_income <= 500000 THEN
        rebate = minimum(12500, tax)
    ELSE
        rebate = 0
    END IF

ELSE IF choice = 2 THEN

    taxable_income = gross_income

    Calculate tax using new-regime slabs

    IF taxable_income <= 1200000 THEN
        rebate = minimum(60000, tax)
    ELSE
        rebate = 0
    END IF

ELSE IF choice = 3 THEN
    DISPLAY "Program terminated"
    STOP

ELSE
    DISPLAY "Invalid choice"
END IF

tax_after_rebate = tax - rebate

health_education_cess = 0.04 * tax_after_rebate

final_tax = tax_after_rebate + health_education_cess
```

DISPLAY all details

STOP

Complete C Program

```
#include <stdio.h>
#include <string.h>

double minimum(double a, double b)
{
    return (a < b) ? a : b;
}

/* Calculate tax under old regime */
double calculateOldTax(double income, int age)
{
    double tax = 0.0;

    if (age < 60)
    {
        if (income <= 250000)
            tax = 0;
        else if (income <= 300000)
            tax = (income - 250000) * 0.05;
        else if (income <= 500000)
            tax = 2500 + (income - 300000) * 0.05;
        else if (income <= 1000000)
            tax = 12500 + (income - 500000) * 0.20;
        else
            tax = 112500 + (income - 1000000) * 0.30;
    }
```

```
    else if (age < 80)
    {
        if (income <= 300000)
            tax = 0;
        else if (income <= 500000)
            tax = (income - 300000) * 0.05;
        else if (income <= 1000000)
            tax = 10000 + (income - 500000) * 0.20;
        else
            tax = 110000 + (income - 1000000) * 0.30;
    }
    else
    {
        if (income <= 500000)
            tax = 0;
        else if (income <= 1000000)
            tax = (income - 500000) * 0.20;
        else
            tax = 100000 + (income - 1000000) * 0.30;
    }

    return tax;
}

/* Calculate tax under new regime */
double calculateNewTax(double income)
{
    double tax = 0.0;

    if (income <= 400000)
        tax = 0;
```

```
    else if (income <= 800000)
        tax = (income - 400000) * 0.05;

    else if (income <= 1200000)
        tax = 20000 + (income - 800000) * 0.10;

    else if (income <= 1600000)
        tax = 60000 + (income - 1200000) * 0.15;

    else if (income <= 2000000)
        tax = 120000 + (income - 1600000) * 0.20;

    else if (income <= 2400000)
        tax = 200000 + (income - 2000000) * 0.25;

    else
        tax = 300000 + (income - 2400000) * 0.30;

    return tax;
}

int main()
{
    char name[100];
    int age, choice;

    double gross_income;
    double standard_deduction;
    double deduction_savings;
    double deduction_pension;
    double deduction_medical;
    double home_loan_interest;
```

```
double savings_account_interest;
double senior_citizen_interest;

double total_deductions;
double taxable_income;
double tax;
double rebate;
double tax_after_rebate;
double health_education_cess;
double final_tax;

printf("Enter taxpayer's name: ");
scanf(" %[^\n]", name);

printf("Enter age: ");
scanf("%d", &age);

printf("Enter gross income: ");
scanf("%lf", &gross_income);

do
{
   printf("\n========== TAX MENU ==========\n");
   printf("1. Old Tax Regime\n");
   printf("2. New Tax Regime\n");
   printf("3. Exit\n");
   printf("Enter your choice: ");
   scanf("%d", &choice);

   if (choice == 1)
   {
      printf("\n--- OLD TAX REGIME ---\n");
```

```
printf("Enter standard deduction: ");
scanf("%lf", &standard_deduction);

printf("Enter deduction savings: ");
scanf("%lf", &deduction_savings);

printf("Enter deduction pension: ");
scanf("%lf", &deduction_pension);

printf("Enter deduction medical: ");
scanf("%lf", &deduction_medical);

printf("Enter home loan interest: ");
scanf("%lf", &home_loan_interest);

printf("Enter savings account interest: ");
scanf("%lf", &savings_account_interest);

printf("Enter senior citizen interest: ");
scanf("%lf", &senior_citizen_interest);

/* Apply maximum limits */
standard_deduction =
    minimum(standard_deduction, 50000);

deduction_savings =
    minimum(deduction_savings, 150000);

deduction_pension =
    minimum(deduction_pension, 50000);
```

```
deduction_medical =
    minimum(deduction_medical, 25000);

home_loan_interest =
    minimum(home_loan_interest, 200000);

savings_account_interest =
    minimum(savings_account_interest, 10000);

senior_citizen_interest =
    minimum(senior_citizen_interest, 50000);

total_deductions =
    standard_deduction +
    deduction_savings +
    deduction_pension +
    deduction_medical +
    home_loan_interest +
    savings_account_interest +
    senior_citizen_interest;

taxable_income =
    gross_income - total_deductions;

if (taxable_income < 0)
    taxable_income = 0;

tax = calculateOldTax(taxable_income, age);

/* Old regime rebate */
if (taxable_income <= 500000)
    rebate = minimum(12500, tax);
```

```
    else
        rebate = 0;

    tax_after_rebate = tax - rebate;

    health_education_cess =
        0.04 * tax_after_rebate;

    final_tax =
        tax_after_rebate + health_education_cess;

    printf("\n=========== TAX DETAILS
===========\n");
    printf("Name                    : %s\n", name);
    printf("Age                     : %d\n", age);
    printf("Gross Income            : Rs. %.2lf\n",
           gross_income);
    printf("Total Deductions        : Rs. %.2lf\n",
           total_deductions);
    printf("Taxable Income          : Rs. %.2lf\n",
           taxable_income);
    printf("Income Tax Before Rebate    : Rs. %.2lf\n",
           tax);
    printf("Rebate                  : Rs. %.2lf\n",
           rebate);
    printf("Income Tax After Rebate     : Rs. %.2lf\n",
           tax_after_rebate);
    printf("Health & Education Cess     : Rs. %.2lf\n",
           health_education_cess);
    printf("Final Tax Payable       : Rs. %.2lf\n",
           final_tax);
}
```

```
else if (choice == 2)
{
    printf("\n--- NEW TAX REGIME ---\n");

    taxable_income = gross_income;

    tax = calculateNewTax(taxable_income);

    /* New regime rebate */
    if (taxable_income <= 1200000)
        rebate = minimum(60000, tax);
    else
        rebate = 0;

    tax_after_rebate = tax - rebate;

    health_education_cess =
        0.04 * tax_after_rebate;

    final_tax =
        tax_after_rebate + health_education_cess;

    printf("\n=========== TAX DETAILS
===========\n");
    printf("Name                    : %s\n", name);
    printf("Age                     : %d\n", age);
    printf("Gross Income            : Rs. %.2lf\n",
           gross_income);
    printf("Taxable Income          : Rs. %.2lf\n",
           taxable_income);
    printf("Income Tax Before Rebate    : Rs. %.2lf\n",
```

```
            tax);
        printf("Rebate                    : Rs. %.2lf\n",
            rebate);
        printf("Income Tax After Rebate   : Rs. %.2lf\n",
            tax_after_rebate);
        printf("Health & Education Cess   : Rs. %.2lf\n",
            health_education_cess);
        printf("Final Tax Payable         : Rs. %.2lf\n",
            final_tax);
    }

    else if (choice == 3)
    {
        printf("\nProgram terminated.\n");
    }

    else
    {
        printf("\nInvalid choice. Please try again.\n");
    }

  } while (choice != 3);

  return 0;
}
```

Test Case Table — Income Tax Program

TC	Age	Gross Income	Regime	Important Input	Expected Result
TC01	30	₹4,00,000	Old	Deductions ₹50,000	Taxable income ₹3,50,000
TC02	30	₹5,00,000	Old	Deductions ₹50,000	Rebate applicable
TC03	45	₹8,00,000	Old	Deductions ₹1,00,000	Tax calculated using old slabs
TC04	65	₹9,00,000	Old	Deductions ₹1,00,000	Senior citizen slab applied
TC05	82	₹12,00,000	Old	Deductions ₹1,00,000	Super senior slab applied
TC06	30	₹8,00,000	New	No deductions	Tax calculated using

TC	Age	Gross Income	Regime	Important Input	Expected Result
					new slabs
TC07	35	₹12,00,000	New	No deductions	Rebate applicable
TC08	40	₹15,00,000	New	No deductions	15% slab applicable
TC09	50	₹25,00,000	New	No deductions	30% slab applicable
TC10	30	₹5,00,000	Invalid choice	—	Invalid choice message

Sample Execution — New Regime

Input

Enter taxpayer's name: Rahul

Enter age: 30

Enter gross income: 800000

========== TAX MENU ==========

1. Old Tax Regime
2. New Tax Regime
3. Exit
Enter your choice: 2

Output

```
--- NEW TAX REGIME ---

========== TAX DETAILS ==========
Name                       : Rahul
Age                        : 30
Gross Income               : Rs. 800000.00
Taxable Income             : Rs. 800000.00
Income Tax Before Rebate   : Rs. 20000.00
Rebate                     : Rs. 20000.00
Income Tax After Rebate    : Rs. 0.00
Health & Education Cess    : Rs. 0.00
Final Tax Payable          : Rs. 0.00
```

Compilation and Execution

Save the program as:

```
income_tax.c
```

Compile:

```
gcc income_tax.c -o income_tax
```

Run:

```
./income_tax
```

For the quadratic program:

```
gcc quadratic.c -o quadratic -lm
```

Run:

```
./quadratic
```

Result

Thus, the C programs for finding the roots of a quadratic equation and calculating income tax under the old and new tax regimes were designed, implemented, compiled, executed, and tested successfully using suitable test cases.

C Program

```
#include <stdio.h>
#include <math.h>

int main()
{
    double a, b, c, D;
    double root1, root2;
    double realPart, imagPart;

    printf("Enter coefficients a, b and c: ");
    scanf("%lf %lf %lf", &a, &b, &c);

    if (a == 0)
    {
        printf("Not a quadratic equation.\n");
    }
    else
    {
        D = b * b - 4 * a * c;

        printf("Discriminant = %.2lf\n", D);

        if (D > 0)
        {
            root1 = (-b + sqrt(D)) / (2 * a);
            root2 = (-b - sqrt(D)) / (2 * a);
```

```
            printf("Two real and distinct roots.\n");
            printf("Root 1 = %.2lf\n", root1);
            printf("Root 2 = %.2lf\n", root2);
        }
        else if (D == 0)
        {
            root1 = -b / (2 * a);
            root2 = root1;

            printf("Two real and equal roots.\n");
            printf("Root 1 = %.2lf\n", root1);
            printf("Root 2 = %.2lf\n", root2);
        }
        else
        {
            realPart = -b / (2 * a);
            imagPart = sqrt(-D) / fabs(2 * a);

            printf("Two imaginary/complex roots.\n");
            printf("Root 1 = %.2lf + %.2lfi\n",
                   realPart, imagPart);
            printf("Root 2 = %.2lf - %.2lfi\n",
                   realPart, imagPart);
        }
    }

    return 0;
}
```

Compilation

```
gcc quadratic.c -o quadratic -lm
```

Execution

```
./quadratic
```

Sample Output — D > 0

Enter coefficients a, b and c: 1 -5 6

Discriminant = 1.00

Two real and distinct roots.

Root 1 = 3.00

Root 2 = 2.00

Sample Output — D = 0

Enter coefficients a, b and c: 1 -4 4

Discriminant = 0.00

Two real and equal roots.

Root 1 = 2.00

Root 2 = 2.00

Sample Output — D < 0

Enter coefficients a, b and c: 1 2 5

Discriminant = -16.00

Two imaginary/complex roots.

Root 1 = -1.00 + 2.00i

Root 2 = -1.00 - 2.00i

---

2. Menu-Driven C Program to Calculate Income Tax

Aim

Write a menu-driven C program to calculate income tax under the **old and new tax regimes in India.**

The program accepts:

- Taxpayer's name
- Age
- Gross income
- Eligible deductions

and calculates the final tax payable.

(a) Calculation of Tax in Old Regime

Input

The program accepts:
standard deduction
deduction savings
deduction pension
deduction medical
home loan interest
savings account interest
senior citizen interest

Maximum Deduction Limits

Deduction	Maximum Limit
Standard deduction	₹50,000
Deduction savings	₹1,50,000
Deduction pension	₹50,000
Deduction medical	₹25,000
Home loan interest	₹2,00,000
Savings account interest	₹10,000
Senior citizen interest	₹50,000

Each deduction should be restricted to its maximum permitted limit.

Calculate Total Deduction and Taxable Income

total_deductions = sum of allowed deductions

taxable_income = gross_income - total_deductions

If:

taxable_income < 0

then:

taxable_income = 0

Old Regime Tax Slabs

Taxable Income Slab	Non-Senior Citizen (<60 years)	Senior Citizen (60 to <80 years)	Super Senior Citizen (>=80 years)
Up to ₹2,50,000	Nil	Nil	Nil
₹2,50,001 to ₹3,00,000	5%	Nil	Nil
₹3,00,001 to ₹5,00,000	5%	5%	Nil
₹5,00,001 to ₹10,00,000	20%	20%	20%
Above ₹10,00,000	30%	30%	30%

Old Regime Tax Rebate

Taxable Income Limit	Rebate
≤ ₹5,00,000	Minimum of ₹12,500 and tax calculated above
> ₹5,00,000	₹0

(b) Calculation of Tax in New Regime

Note: The image labels this section as "(b) Calculation of Tax in Old regime", but the slab table shown is clearly the **new regime** slab structure. So it should be written as **New Regime**.

Calculate Taxable Income

Under the new regime:

taxable_income = gross_income

No deductions from the old-regime section are applied here.

New Regime Tax Slabs

Taxable Income Slab	Tax Rate
Up to ₹4,00,000	Nil
₹4,00,001 to ₹8,00,000	5%
₹8,00,001 to ₹12,00,000	10%
₹12,00,001 to ₹16,00,000	15%
₹16,00,001 to ₹20,00,000	20%
₹20,00,001 to ₹24,00,000	25%
Above ₹24,00,000	30%

New Regime Tax Rebate

Taxable Income Limit	Rebate
≤ ₹12,00,000	Minimum of ₹60,000 and tax calculated above
> ₹12,00,000	₹0

(c) Final Calculations

These calculations are common to both regimes.

tax_after_rebate = tax - rebate

health_education_cess = 4% of tax_after_rebate

final_tax = tax_after_rebate + health_education_cess

(d) Output

The program should display:

Name

Age

Gross Income

Taxable Income

Income Tax Before Rebate

Rebate

Income Tax After Rebate

Health & Education Cess

Final Tax Payable

Algorithm — Income Tax Program

1. Start.
2. Input taxpayer's name.
3. Input taxpayer's age.
4. Input gross income.
5. Display menu:
    - Old Tax Regime
    - New Tax Regime
    - Exit
6. If the user selects the old regime:
    - Input all deductions.
    - Apply the maximum limits to the deductions.
    - Calculate total deductions.
    - Calculate taxable income.
    - If taxable income is negative, set it to zero.
    - Determine the applicable tax slabs according to age.

- Calculate income tax.
- Calculate old-regime rebate.

7. If the user selects the new regime:
   - Set taxable income equal to gross income.
   - Calculate tax according to the new-regime slabs.
   - Calculate new-regime rebate.
8. Calculate tax after rebate.
9. Calculate 4% Health and Education Cess.
10. Calculate final tax payable.
11. Display all details.
12. If Exit is selected, terminate the program.
13. Stop.

Pseudocode

```
START

INPUT name
INPUT age
INPUT gross_income

DISPLAY MENU

1. Old Tax Regime
2. New Tax Regime
3. Exit

INPUT choice

IF choice = 1 THEN

    INPUT all deductions
```

```
    Limit each deduction according to maximum allowed limit

    total_deductions =
        standard deduction +
        deduction savings +
        deduction pension +
        deduction medical +
        home loan interest +
        savings account interest +
        senior citizen interest

    taxable_income = gross_income - total_deductions

    IF taxable_income < 0 THEN
        taxable_income = 0
    END IF

    Determine age category

    Calculate tax using old-regime slabs

    IF taxable_income <= 500000 THEN
        rebate = minimum(12500, tax)
    ELSE
        rebate = 0
    END IF

ELSE IF choice = 2 THEN

    taxable_income = gross_income

    Calculate tax using new-regime slabs
```

```
    IF taxable_income <= 1200000 THEN
        rebate = minimum(60000, tax)
    ELSE
        rebate = 0
    END IF

ELSE IF choice = 3 THEN
    DISPLAY "Program terminated"
    STOP

ELSE
    DISPLAY "Invalid choice"
END IF

tax_after_rebate = tax - rebate

health_education_cess = 0.04 * tax_after_rebate

final_tax = tax_after_rebate + health_education_cess

DISPLAY all details

STOP
```

Complete C Program

```c
#include <stdio.h>
#include <string.h>

double minimum(double a, double b)
{
    return (a < b) ? a : b;
```

```
}

/* Calculate tax under old regime */
double calculateOldTax(double income, int age)
{
    double tax = 0.0;

    if (age < 60)
    {
        if (income <= 250000)
            tax = 0;
        else if (income <= 300000)
            tax = (income - 250000) * 0.05;
        else if (income <= 500000)
            tax = 2500 + (income - 300000) * 0.05;
        else if (income <= 1000000)
            tax = 12500 + (income - 500000) * 0.20;
        else
            tax = 112500 + (income - 1000000) * 0.30;
    }
    else if (age < 80)
    {
        if (income <= 300000)
            tax = 0;
        else if (income <= 500000)
            tax = (income - 300000) * 0.05;
        else if (income <= 1000000)
            tax = 10000 + (income - 500000) * 0.20;
        else
            tax = 110000 + (income - 1000000) * 0.30;
    }
    else
```

```
    {
        if (income <= 500000)
            tax = 0;
        else if (income <= 1000000)
            tax = (income - 500000) * 0.20;
        else
            tax = 100000 + (income - 1000000) * 0.30;
    }

    return tax;
}

/* Calculate tax under new regime */
double calculateNewTax(double income)
{
    double tax = 0.0;

    if (income <= 400000)
        tax = 0;

    else if (income <= 800000)
        tax = (income - 400000) * 0.05;

    else if (income <= 1200000)
        tax = 20000 + (income - 800000) * 0.10;

    else if (income <= 1600000)
        tax = 60000 + (income - 1200000) * 0.15;

    else if (income <= 2000000)
        tax = 120000 + (income - 1600000) * 0.20;
```

```
    else if (income <= 2400000)
        tax = 200000 + (income - 2000000) * 0.25;

    else
        tax = 300000 + (income - 2400000) * 0.30;

    return tax;
}

int main()
{
    char name[100];
    int age, choice;

    double gross_income;
    double standard_deduction;
    double deduction_savings;
    double deduction_pension;
    double deduction_medical;
    double home_loan_interest;
    double savings_account_interest;
    double senior_citizen_interest;

    double total_deductions;
    double taxable_income;
    double tax;
    double rebate;
    double tax_after_rebate;
    double health_education_cess;
    double final_tax;

    printf("Enter taxpayer's name: ");
```

```
scanf(" %[^\n]", name);

printf("Enter age: ");
scanf("%d", &age);

printf("Enter gross income: ");
scanf("%lf", &gross_income);

do
{
    printf("\n========== TAX MENU ==========\n");
    printf("1. Old Tax Regime\n");
    printf("2. New Tax Regime\n");
    printf("3. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    if (choice == 1)
    {
        printf("\n--- OLD TAX REGIME ---\n");

        printf("Enter standard deduction: ");
        scanf("%lf", &standard_deduction);

        printf("Enter deduction savings: ");
        scanf("%lf", &deduction_savings);

        printf("Enter deduction pension: ");
        scanf("%lf", &deduction_pension);

        printf("Enter deduction medical: ");
        scanf("%lf", &deduction_medical);
```

```
printf("Enter home loan interest: ");
scanf("%lf", &home_loan_interest);

printf("Enter savings account interest: ");
scanf("%lf", &savings_account_interest);

printf("Enter senior citizen interest: ");
scanf("%lf", &senior_citizen_interest);

/* Apply maximum limits */
standard_deduction =
    minimum(standard_deduction, 50000);

deduction_savings =
    minimum(deduction_savings, 150000);

deduction_pension =
    minimum(deduction_pension, 50000);

deduction_medical =
    minimum(deduction_medical, 25000);

home_loan_interest =
    minimum(home_loan_interest, 200000);

savings_account_interest =
    minimum(savings_account_interest, 10000);

senior_citizen_interest =
    minimum(senior_citizen_interest, 50000);
```

```
total_deductions =
    standard_deduction +
    deduction_savings +
    deduction_pension +
    deduction_medical +
    home_loan_interest +
    savings_account_interest +
    senior_citizen_interest;

taxable_income =
    gross_income - total_deductions;

if (taxable_income < 0)
    taxable_income = 0;

tax = calculateOldTax(taxable_income, age);

/* Old regime rebate */
if (taxable_income <= 500000)
    rebate = minimum(12500, tax);
else
    rebate = 0;

tax_after_rebate = tax - rebate;

health_education_cess =
    0.04 * tax_after_rebate;

final_tax =
    tax_after_rebate + health_education_cess;
```

```
        printf("\n=========== TAX DETAILS
===========\n");
        printf("Name                    : %s\n", name);
        printf("Age                     : %d\n", age);
        printf("Gross Income            : Rs. %.2lf\n",
            gross_income);
        printf("Total Deductions        : Rs. %.2lf\n",
            total_deductions);
        printf("Taxable Income          : Rs. %.2lf\n",
            taxable_income);
        printf("Income Tax Before Rebate    : Rs. %.2lf\n",
            tax);
        printf("Rebate                  : Rs. %.2lf\n",
            rebate);
        printf("Income Tax After Rebate     : Rs. %.2lf\n",
            tax_after_rebate);
        printf("Health & Education Cess     : Rs. %.2lf\n",
            health_education_cess);
        printf("Final Tax Payable       : Rs. %.2lf\n",
            final_tax);
    }

    else if (choice == 2)
    {
        printf("\n--- NEW TAX REGIME ---\n");

        taxable_income = gross_income;

        tax = calculateNewTax(taxable_income);

        /* New regime rebate */
        if (taxable_income <= 1200000)
```

```
        rebate = minimum(60000, tax);
    else
        rebate = 0;

    tax_after_rebate = tax - rebate;

    health_education_cess =
        0.04 * tax_after_rebate;

    final_tax =
        tax_after_rebate + health_education_cess;

    printf("\n=========== TAX DETAILS
===========\n");
    printf("Name                    : %s\n", name);
    printf("Age                     : %d\n", age);
    printf("Gross Income            : Rs. %.2lf\n",
          gross_income);
    printf("Taxable Income          : Rs. %.2lf\n",
          taxable_income);
    printf("Income Tax Before Rebate   : Rs. %.2lf\n",
          tax);
    printf("Rebate                  : Rs. %.2lf\n",
          rebate);
    printf("Income Tax After Rebate    : Rs. %.2lf\n",
          tax_after_rebate);
    printf("Health & Education Cess    : Rs. %.2lf\n",
          health_education_cess);
    printf("Final Tax Payable       : Rs. %.2lf\n",
          final_tax);
}
```

```
        else if (choice == 3)
        {
            printf("\nProgram terminated.\n");
        }

        else
        {
            printf("\nInvalid choice. Please try again.\n");
        }

    } while (choice != 3);

    return 0;
}
```

Test Case Table — Income Tax Program

TC	Age	Gross Income	Regime	Important Input	Expected Result
TC01	30	₹4,00,000	Old	Deductions ₹50,000	Taxable income ₹3,50,000
TC02	30	₹5,00,000	Old	Deductions ₹50,000	Rebate applicable
TC03	45	₹8,00,000	Old	Deductions ₹1,00,000	Tax calculated using

TC	Age	Gross Income	Regime	Important Input	Expected Result
					old slabs
TC04	65	₹9,00,000	Old	Deductions ₹1,00,000	Senior citizen slab applied
TC05	82	₹12,00,000	Old	Deductions ₹1,00,000	Super senior slab applied
TC06	30	₹8,00,000	New	No deductions	Tax calculated using new slabs
TC07	35	₹12,00,000	New	No deductions	Rebate applicable
TC08	40	₹15,00,000	New	No deductions	15% slab applicable

TC	Age	Gross Income	Regime	Important Input	Expected Result
TC09	50	₹25,00,000	New	No deductions	30% slab applicable
TC10	30	₹5,00,000	Invalid choice	—	Invalid choice message

Sample Execution — New Regime

Input

Enter taxpayer's name: Rahul

Enter age: 30

Enter gross income: 800000

=========== TAX MENU ===========

1. Old Tax Regime
2. New Tax Regime
3. Exit

Enter your choice: 2

Output

--- NEW TAX REGIME ---

=========== TAX DETAILS ===========

Name : Rahul

Age : 30

Gross Income : Rs. 800000.00

Taxable Income : Rs. 800000.00

Income Tax Before Rebate : Rs. 20000.00

Rebate : Rs. 20000.00

Income Tax After Rebate : Rs. 0.00

Health & Education Cess : Rs. 0.00

Final Tax Payable : Rs. 0.00

Compilation and Execution

Save the program as:

income_tax.c

Compile:

gcc income_tax.c -o income_tax

Run:

./income_tax